Aeron Jay P. Bulatao

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Experience

Paynamics Technologies Inc. Makati, Philippines (Hybrid)
Software Engineer *Jul 2026 – Present*

- Build and maintain payment gateway APIs and backend services in **Kotlin (Javalin)** and **Java (Spring Boot)** that process merchant transactions.
- Integrate new partner payment channels and merchant checkout endpoints, validating every request against its contract and persisting transaction records across **PostgreSQL** and **MongoDB**.
- Strengthen error handling in payment flows so failed or partial transactions are caught, logged and recoverable instead of silently dropped.
- Contribute to internal merchant tools and operational workflows with React/TypeScript and AWS.

Freelance Full-Stack Developer Remote
Solo builds for private clients, from requirements to production *2026*

- **Task Tracker (NDA client):** Built a daily task assignment and verification system used by **390 people every day**, with five roles, proof-of-work submissions, a verdict and resubmission workflow, and graded Excel performance reports (**Next.js, TypeScript, Supabase, Cloudflare Workers**).
- Enforced access for five roles with **Postgres row-level security**; resolved infinite policy recursion with `SECURITY DEFINER` PL/pgSQL helper functions, and moved cross-member views into authorized server actions cached per group.
- Kept the app inside Cloudflare Workers' 10 ms CPU budget and the database egress limit through tag-based cache invalidation, in-browser photo compression and server-side paging; photos stored on **Cloudflare R2**.
- **Event Desk:** Replaced an events office's growing Excel sheet with a **React/TypeScript + Supabase** tracker now in daily use, separating past and upcoming requests with server-side filtering and pagination.
- Automated follow-up with **Deno edge functions**: daily email reminders 5 days, 1 day and same-day before each event (computed in Manila time), one-click Telegram posts with attachments, and nightly **GitHub Actions** database backups to R2 with 7-day retention.

FPD Asia Property Services, Inc. Makati, Philippines (On-site)
Full-Stack Developer & Tech Lead (Internship) *Mar 2026 – May 2026*

- Led end-to-end development of an internal property management web application as Technical Lead and Full-Stack Developer using **React/TypeScript** and **Java/Spring Boot**.
- Designed and deployed event-driven serverless functions on **AWS Lambda** to automate onboarding emails, eliminating manual handling and reducing turnaround time.
- Configured **AWS IAM** roles and least-privilege policies governing interactions between Lambda, Cognito and the Spring Boot backend.
- Integrated **AWS Cognito** for authentication and role-based access control across the application.

Private E-Commerce Client (Canada) Remote
Automation & Web Scraping Engineer (Freelance) *Jan 2025 – Mar 2025*

- Delivered a paid, production multi-site scraping system in **Python/Playwright** covering **7+ competitor sites** including Amazon.ca, automating price research that took **5–8 hours** of manual work every week.
- Built the pipeline end to end: scheduled trigger → headless Chromium with anti-bot measures → dynamic DOM traversal across up to 20 paginated pages per site → structured CSV output.
- Designed one scraper module per site behind a master dispatcher, so new competitors are onboarded without touching working code; added selector fallback chains to survive markup changes.

Projects

AmBOT | *Python, FastAPI, FAISS, Mistral 7B, React, TypeScript* | github.com/aeronjay/AmangBot *2025 – 2026*

- Built a RAG chatbot (team thesis; sole developer of the system) that answers student questions from 225 topic-sized chunks of the university's handbook, charter and manuals, citing its source and declining when the documents don't cover a question.

- Designed layered retrieval: **nomic-embed fine-tuned on 8,432** question–passage pairs, hybrid FAISS + BM25 search, a cross-encoder reranker and a relevance threshold, feeding a locally hosted **Mistral 7B** (llama.cpp, GPU).
- Added Taglish follow-up rewriting, token streaming over server-sent events, an admin PDF ingestion dashboard, and automated evaluation of **84 questions across 250 runs** with BLEU, ROUGE-L and METEOR scoring.

SmartSMS Barangay | *React, Node.js, Express, MongoDB, JWT* | github.com/aeronjay/SmartSMS-Barangay *2025*

- Built a records and services system now used by **Barangay 551, Zone 54, Manila**: targeted SMS announcements to filtered resident groups, households with printable RBI forms, and an audit trail.
- Developed the Express backend (∼**40 REST endpoints**, JWT auth, superadmin role checks) and integrated an **Android phone as a free SMS gateway** with delivery-status tracking, avoiding per-message API costs.
- Implemented email-verified certificate requests with 6-digit, 10-minute codes held in a pending collection, so only verified requests reach the admin queue; deployed on Render.

AI Support Desk for Shopify | *n8n, Gemini API, Shopify GraphQL, Gmail API, Docker* *2026*

- Built a 54-node **n8n** workflow that classifies support emails with Gemini, fetches the real order from the Shopify Admin GraphQL API, applies refund and return rules in code, and saves each reply as a Gmail draft for human approval.
- Added a quality gate that flags risky tickets (low AI confidence, unhappy customers, high-value orders, repeat contacts) to Discord, plus a daily HTML report and an error-alert workflow; found and fixed 4 bugs in end-to-end testing.

Skills & Interests

Languages: Kotlin, Java, TypeScript, JavaScript, Python, SQL

Backend & Frontend: Javalin, Spring Boot, Node.js, Express, FastAPI, REST APIs, React, Next.js, Tailwind CSS

Data: PostgreSQL (row-level security), MongoDB, Supabase, FAISS, BM25

Cloud & DevOps: AWS (Lambda, IAM, Cognito), Cloudflare Workers & R2, Docker, Vercel, Render, GitHub Actions, Git, Linux

AI & Automation: RAG, LangChain, Sentence Transformers, llama.cpp, Gemini API, n8n, Make, Zapier, Playwright

Interests: Chess, tennis, weightlifting

Education

Eulogio "Amang" Rodriguez Institute of Science and Technology (EARIST) Manila, Philippines
Bachelor of Science in Computer Science | GPA: 1.49 (PH scale, ≈ 3.8/4.0) *2026*

- **Thesis:** AmBOT, a retrieval-augmented chatbot that answers student questions from the university's own handbooks and manuals (sole developer of the system).